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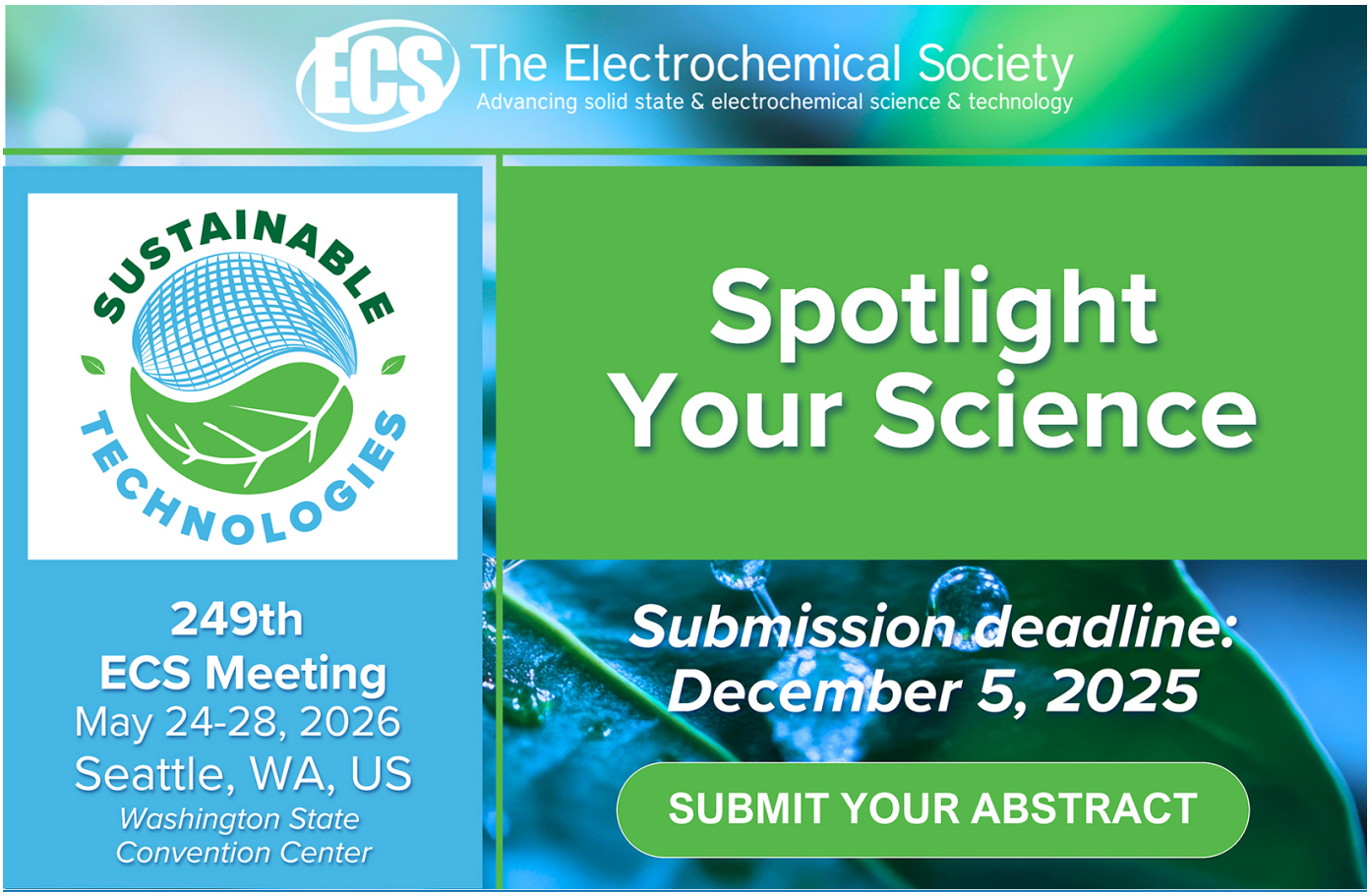
Observation of equiaxed dendrite growth and motion in Al-Cu alloy using 4D-CT

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Observation of equiaxed dendrite growth and motion in Al-Cu alloy using 4D-CT

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Abstract. Time-resolved computed tomography (4D-CT) was performed to observe the formation process of equiaxed grain structure in Al-5mass%Cu alloy refined by inoculant TiB_2 particles. The temporal resolution of as short as 1 s was achieved using a high-brilliance X-ray beam at a beamline BL20B2 of SPring-8 (Hyogo, Japan). A filtering method based on a phase field model was used to improve the quality of reconstructed images, evaluating equiaxed grain motion in conjunction with their dendritic growth. Nucleation events of α-Al was followed by the settlement of equiaxed grains. The moving velocity in the vertical direction was first -100 μm/s and then reduced to zero due to sedimentation at the bottom of the specimen as solidification proceeded. The solid fraction was 0.15 when the observation region was filled with stacked equiaxed grains. No nucleation events occurred in the remaining liquid phase between stacked equiaxed grains. 4D-CT observation demonstrates that the number and arrangement of equiaxed grains in the early stage of solidification determines the grain size distribution in the as-cast microstructure.

1. Introduction

1.1 Time-resolved and in-situ observation of solidification phenomena in three dimensions

Time-resolved computed tomography using synchrotron radiation X-rays (4D-CT) has been developed to elucidate solidification phenomena in metallic materials. Three-dimensional and time-resolved observation provides valuable information related to microstructure evolution during solidification. For example, the late stage of solidification in Al-4mass%Cu alloy was observed using 4D-CT with a white X-ray beam (peak energy: 40 keV), which allows us to improve the quality of the tomographic images owing to an increase in the intensity of transmission X-ray(1). Dendrite arm coarsening in Al-15mass%Cu alloy was observed using a monochromatized X-ray of 30 keV, and the observed microstructure was compared with a phase-field simulation(2). Dendrite morphology during the early stage of solidification in Al-24mass%Cu alloy was reconstructed using a time-interlaced model-based iterative reconstruction method, which improved the temporal resolution(3). The evolution of the primary, intermetallic, and eutectic phases in Al-Si-Cu-Fe alloy (commercial Al-A319 alloy) during solidification was observed(4). More recently, solidification in metallic alloys consisting of higher attenuation elements such as Fe alloys(5, 6), Co alloys(5), Ni alloys(5), CrMnFeCoNi alloys(7), and Zn alloys(8) was successfully

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observed using 4D-CT. In addition, 4D-CT observation unveiled transgranular liquation cracking of solid grains in semisolid Al-Cu alloy(9). Translational and rotational motion of solid grains during compression of semisolid Al-10mass%Cu alloy was evaluated using a combination technique of 4D-CT and three-dimensional X-ray diffraction (3DXRD) for determining crystallographic orientations of solid grains(10-12). Thus, 4D-CT is becoming an indispensable tool for directly understanding solidification phenomena.

4D-CT is, however, subject to a fundamental compromise between spatial and temporal resolution. During 4D-CT observation, an X-ray beam is irradiated to a rotating specimen, and projection images (X-ray transmission images) are sequentially captured. Image reconstruction is performed using each projection image dataset, including images captured during 180-degree rotation. The time duration for capturing each dataset (i.e., rotation speed) indicates temporal resolution, and the pixel size of the projection image (the voxel size in the tomographic image) indicates spatial resolution. The number of projection images in each dataset links to the quality of tomographic images, affecting the spatial resolution. Thus, the observation condition has been limited even in the experimental setup using high-brilliance synchrotron radiation X-rays.

The conditions for the reported 4D-CT observation are summarized in Table 1. In previous studies, the rotation speed was 0.25 rps(4 s for 360-degree rotation) at most to ensure the sufficient quality of the tomographic images. This study is motivated to enhance the temporal resolution of 4D-CT observation by optimizing capturing circumstances and utilizing suitable image processing for solidification microstructure.

Table 1. Reported and present 4D-CT condition for observation of solidification in metallic alloys.

Composition(mass%)	Specimen diameter (mm)	Rotation speed (rps)	Temporal resolution (s)	Projection image	Reference
Al-4Cu[a]	1	0.1	10	500	Ludwig(1)
Al-15Cu	2	0.003	346	721	Aagesen(2)
Al-24Cu	1	0.05	18	2000	Gibbs(3)
Al-7.52Si-3.53Cu-0.5Fe	2.5	0.25	4	1001	Puncreobutr (4)
Ni-14Hf, Fe-11Hf, Co-18Hf	1.4	0.005	900	90	Azeem(5)
Fe-0.45C-0.6Mn-2Si	0.8	0.25	4	200	Yasuda(6)
CrMnFeCoNi (equimolar)	1	0.25	4	400	Nakano(7)
Zn-4Al	0.7	0.5	2	200	Yasuda(8)
Al-15Cu[b]	2	2	0.5	200	Narumi(17)
Al-5Cu	0.8	1	1	240	Present study

[a] White X-ray (maximum energy: 40keV). [b] Pink beam (White X-ray whose X-ray energy higher than 30keV was cut by a mirror).

1.2 Growth and motion of equiaxed grains in molten alloy

Equiaxed grains are often observed in the casting process. They are formed in the melt owing to fragmentation and detachment of existing columnar dendrite arms and heterogeneous nucleation on inoculated nucleation agents. Motion (i.e., floating and settlement) of equiaxed grains due to melt flow and buoyancy force is associated with latent heat release and solute transport, affecting the thermal and concentration fields surrounding equiaxed grains(13). Furthermore, their sedimentation gives rise to mechanical interaction with each other. In such a semisolid region, shear deformation can be localized, forming segregation bands and porosity(14). In this respect, it is insufficient only to understand the dendritic growth of equiaxed grains for elucidating the formation of equiaxed grain structure and casting defects. The motion of equiaxed grains in the melt and subsequent sedimentation must be considered. If the moving velocity of equiaxed grains is more than 10^1 μm/s, the rotation rate more than 0.5 rps (2 s for 360-degree-rotation) is required to detect their motion using 4D-CT with the voxel size of 5 μm.

4D-CT observation also enables the evaluation of semisolid microstructure due to the sedimentation of equiaxed grains, which helps understand the deformation mechanism of semisolid alloy. Specifically, the coherency point, which is the solid fraction where equiaxed grains form a network to propagate applied force and exhibit mechanical interaction, can be measured directly. The reported coherency points vary depending on the measurement method. In thermal analysis and rheological measurements, the coherency point ranged from 0.1 to 0.3(15). In other measurements, the coherency point depended on the structure: 0.1 with coarse grains and 0.5 with fine grains(15). There is a report that the coherency point was 0.3-0.4, according to the observation of the structure in the sedimentation layer of equiaxed grains(16). 4D-CT observation will provide a specific solid fraction corresponding to the coherency point.

This study aims to achieve high-speed 4D-CT with temporal resolution shorter than 2 s and to analyse the formation process of equiaxed grain structure in Al-Cu alloys. 4D-CT with a temporal resolution of 0.5 s was preliminary examined to observe equiaxed solidification in Al-15mass%Cu alloy(17). The temporal resolution and X-ray absorption contrast between the solid and liquid phases were enough to evaluate the motion of equiaxed grains and their influence on the grain structure evolution. On the contrary, this study focuses on evaluating the growth and motion of equiaxed grains in the melt during solidification in Al-Cu alloy close to the practical composition in 2xxx Al alloy (4-6 mass%Cu). Such composition is more challenging because of the smaller X-ray absorption contrast between the solid and liquid phases than the previous study(17), leading to deterioration of the quality of reconstructed images due to a smaller signal-to-noise ratio. We report the effect of equiaxed grain motion on the formation of equiaxed grain structure and the solid fraction when equiaxed grains are stacked and have mechanical contact with each other, corresponding to the coherency point.

2. Experimental procedure

4D-CT observations were performed at the imaging beamline BL20B2 in SPring-8 (Synchrotron radiation facility, Hyogo, Japan). BL20B2 provides an X-ray beam of 40 keV monochromatized by a double multilayer monochromator, enabling us to perform fast tomography with a sufficient field of view and spatial resolution for observing microstructure evolution in metallic materials.

Figure 1 shows a schematic illustration of the setup of 4D-CT. The details of the setup are given elsewhere(17). Rod-shaped Al-5mass%Cu alloy inoculated with TiB_2 was inserted into a high-purity alumina tube (inner diameter: 0.8 mm) and heated under vacuum (<1 Pa). The specimen was cooled at a cooling rate of 30 K/min from its complete molten state. The incident

beam (width: 5 mm, height: 3 mm) was used to capture the projection images of the specimen rotating at 1 rps.

The 3D image was sequentially reconstructed using 240 projection images over a 180-degree rotation by the convolution back-projection method. The voxel size of the reconstructed images was 5.5 μm x 5.5 μm x 5.5 μm. For image processing of the reconstructed 3D images, a filter based on a phase field model (referred to as phase field filter, PFF) was used. In PFF, the solid fraction is first calculated from the brightness distribution in the reconstructed image. The solid-liquid interface is moved through the calculation by phase field method with appropriate parameters so that the solid fraction obtained from the calculation by phase field method converges to the solid fraction obtained from the brightness distribution in the reconstructed image. Consequently, the curvature effect acts as the driving force for moving the interface, and noise is sufficiently removed. Further details of PFF are given elsewhere(6-8).

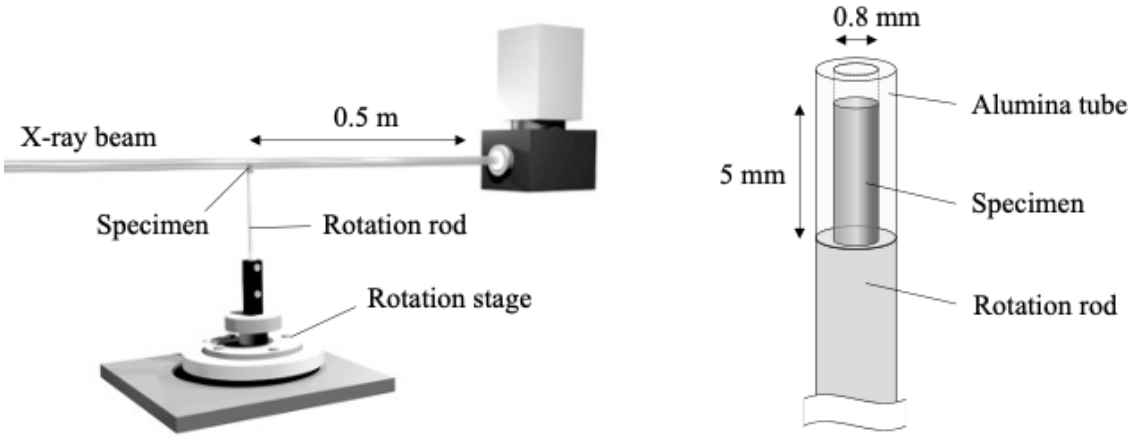


Figure 1. Setup for 4D-CT observation(17).

3. Results and discussion

Fig. 2(a) shows a reconstructed image of Al-5mass%Cu alloy (diameter: 0.8 mm) cooled at a cooling rate of 30 K/min. The periphery of the specimen contained refraction contrast in the transmission images, and the effect of noise was significant; thus, a region with a diameter of 0.6 mm from the centre was extracted. When the X-ray energy is set to 40 keV, the mass absorption coefficient of Al is about one-tenth that of Cu. Despite the higher density of the solid phase compared to the liquid phase, the X-ray absorption of the liquid phase exceeds that of the solid phase in Al-5mass%Cu alloy. Thus, the dark and bright regions are the solid and liquid phases. Although the brightness distribution in the reconstructed image shows that equiaxed grains exist in Fig. 2(a), accurate tracing of the solid-liquid interface is difficult, and consequently, it is impossible to extract the equiaxed grains and track their motion.

Using the reconstructed images, we evaluated the solid fraction as a function of solidification time, as shown in Fig. 3. The distribution of brightness in the reconstructed images, which are proportional to the linear X-ray absorption coefficient, was fitted by two different Gaussian functions: one is the liquid phase, and the other is the solid phase. The calculated solid fraction was used as a constraint condition in PFF(6-8).

The solid fraction calculated from the brightness distribution of Fig. 2(a) was 0.09. Fig. 2 (b) shows an image after processed by PFF using the solid fraction (0.09) in Fig. 2 (a) as a constraint condition and a driving force that the solid fraction obtained from calculation by phase field method converges to 0.09. Here, the phase-field parameters (φ) for the solid and liquid phases are defined as 1 and 0. The solid-liquid interface shape and dendrite arms are confirmed as PFF sufficiently removed the noise. The solid fraction as a function of solidification time, which was obtained from PFF, is also shown in Fig. 3. The increase in solid fractions with solidification time obtained from the reconstructed images and the PFF are consistent and show a tendency commonly observed in specimens solidified at a constant cooling rate.

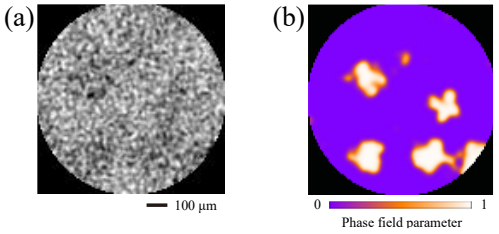


Figure 2. (a) Reconstructed image of equiaxed structure in Al-5mass%Cu alloy (30 K/min); (b) Slice image after image processing based on the phase-field model.

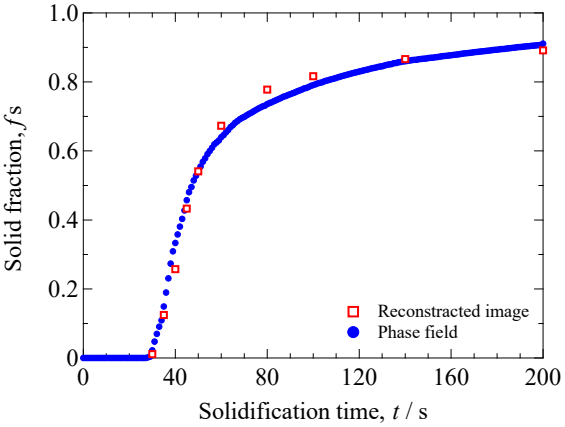


Figure 3. Solid fraction as a function of solidification time.

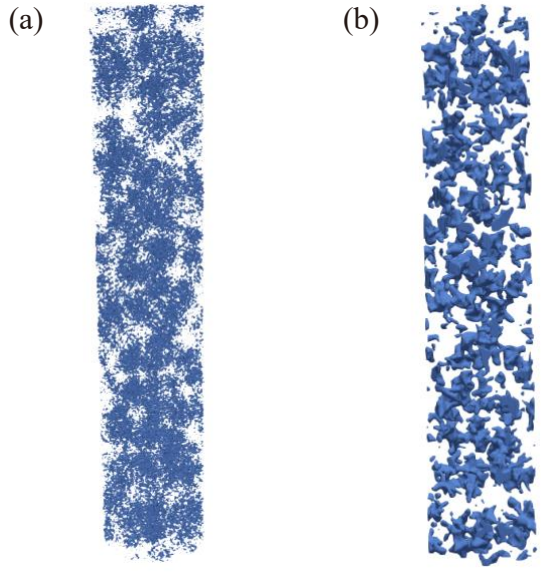


Figure 4. Three-dimensional image (a) before and (b) after image processing based on the phase-field model.

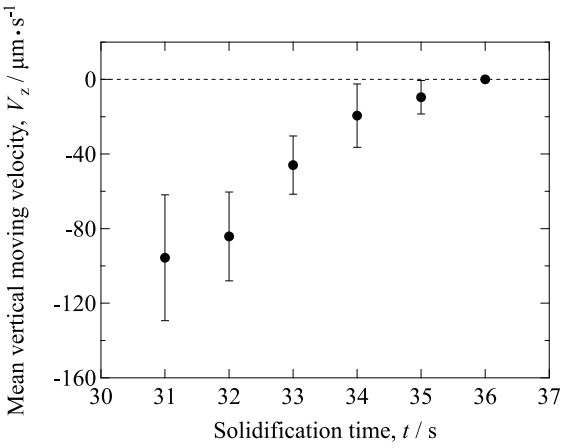


Figure 5. Mean vertical moving velocity as a function of solidification time.

Fig. 4 shows 3D images obtained from the dataset that includes Fig. 1. The observation region is 0.6 mm in diameter and 3.1 mm in height. Using time-resolved data, PFF allows extracting the solid phase from the 3D image and tracking the equiaxed grains. Fig. 5 shows the mean moving velocity in the vertical direction of equiaxed grains as a function of the solidification time. The error bar indicates the standard deviation. In the early stage of solidification, equiaxed grains nucleated in the melt and then settled at -100 μm/s, accumulating at the bottom of the observation region. The travelling distance of the equiaxed grains during 180-degree rotation (0.5 s) corresponds to 5-10 times the voxel size; thus, the shape of the equiaxed grains in the reconstructed images includes the effects of their motion. When Stokes equation was applied with an assumption of spherical grains with 100 μm in diameter, the moving velocity was calculated as -490 μm/s. Thus, the shape of equiaxed grains could be deviated from the sphere even in the early stage of solidification. As solidification proceeded, the moving velocity approached zero. The observation region was filled with stacked equiaxed grains when the solid fraction became 0.15, which is smaller than the close-packed fraction of spheres (0.74). Including the fact that equiaxed grains deviate from the spherical shape, the sedimentation region was less dense than the closest packing. In the sedimentation region, the development of the primary and secondary arms was detected rather than nucleation of another grain. In other words, the grain size in the as-cast microstructure is determined by the number and arrangement of the equiaxed grains when they accumulate during solidification.

4. Summary

Three-dimensional and time-resolved observation of solidification in Al-5mass%Cu alloy inoculated with TiB_2 was performed using 4D-CT. Temporal resolution as short as 1 s caused significant noise in the reconstructed images. Applying PFF, the quality of the reconstructed images was significantly improved, enabling the evaluation of the growth and motion of equiaxed grains. In the early stage of solidification, the settlement of equiaxed grains was detected. The moving velocity in the vertical direction was -100 μm/s, which was one-fifth of that estimated by the Stokes equation. The moving velocity became smaller as solidification proceeded. Equiaxed grains were completely stacked in the observation region when the solid fraction reached 0.15. Once the sedimentation was completed, no nucleation event occurred in the remaining liquid phase. This study demonstrated that the number and arrangement of equiaxed grains in the early stage of solidification determine the final grain size distribution in as-cast microstructure.

As demonstrated above, 4D-CT with PFF would provide beneficial information for building the three-dimensional model of equiaxed solidification. We expect that the model will be useful for understanding the formation of an equiaxed grain structure and predicting defect formation quantitatively.

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